CASPER

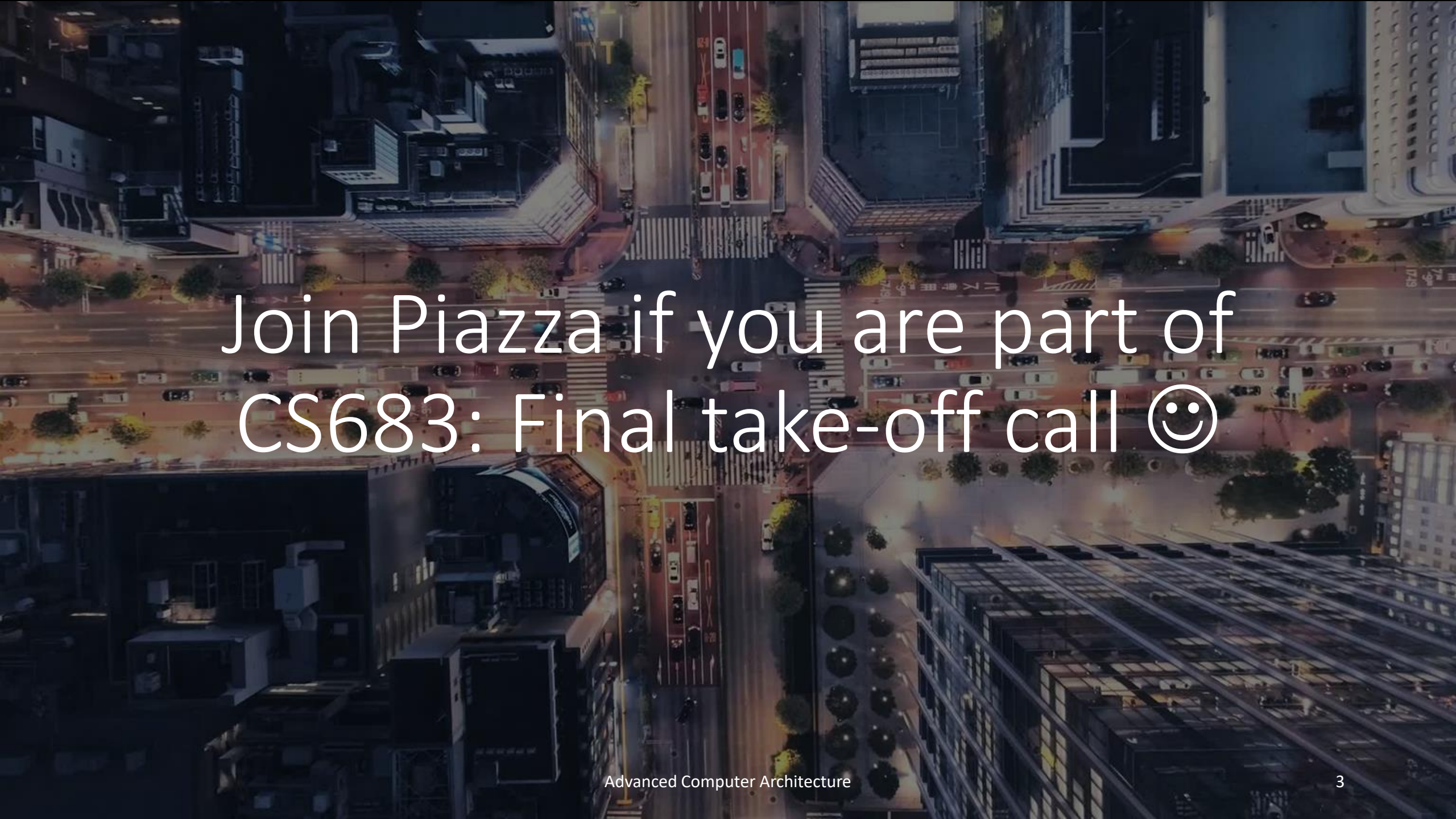
CS683: Advanced Computer Architecture-2024

L3-Beyond Cache and beyond …….

https://www.cse.iitb.ac.in/~biswa/courses.html

https://www.cse.iitb.ac.in/~biswa/

Phones (smart/non-smart) on silence plz, Thanks


Join Piazza if you are part of
CS683: Final take-off call ☺

Summary of Lecture-2

Confused?

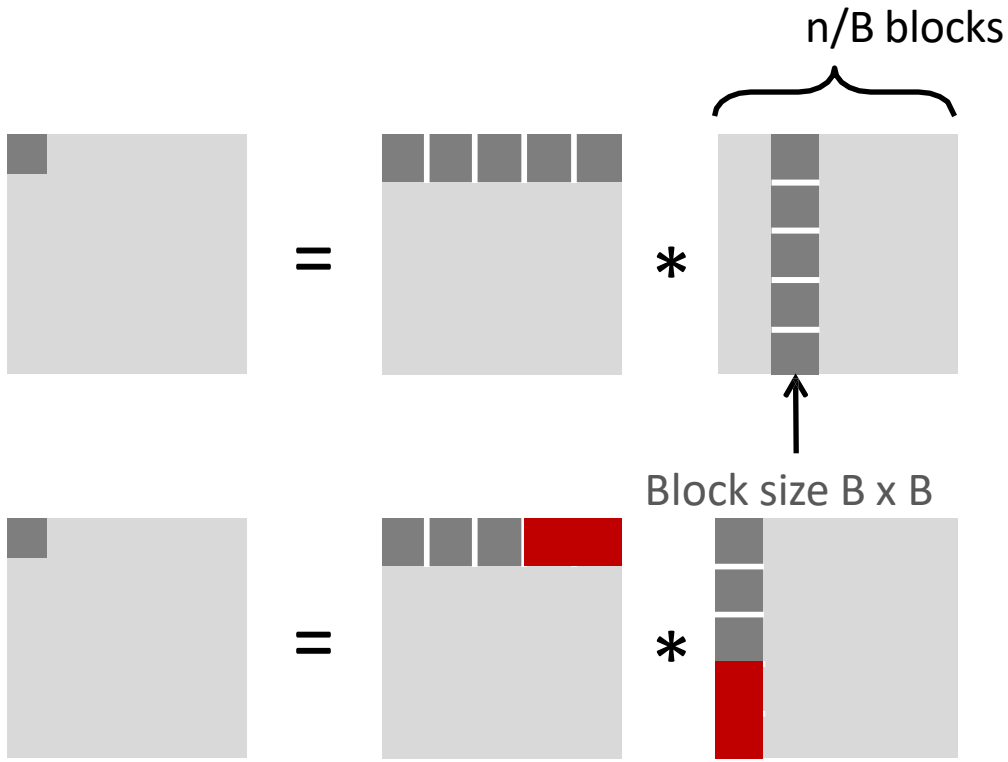
Cache Miss Analysis

- Assume:
  - Cache block = 8 doubles
  - Cache size << n (much smaller than n)
  - Three blocks ■ fit into cache, i.e., $3B^2 < C$

- First (block) iteration:
  - $\frac{B^2}{8}$ misses for each block (tile)
  - $2 * \frac{n}{B} * \frac{B^2}{8} = \frac{2*nB}{8}$

  (ignoring matrix C)

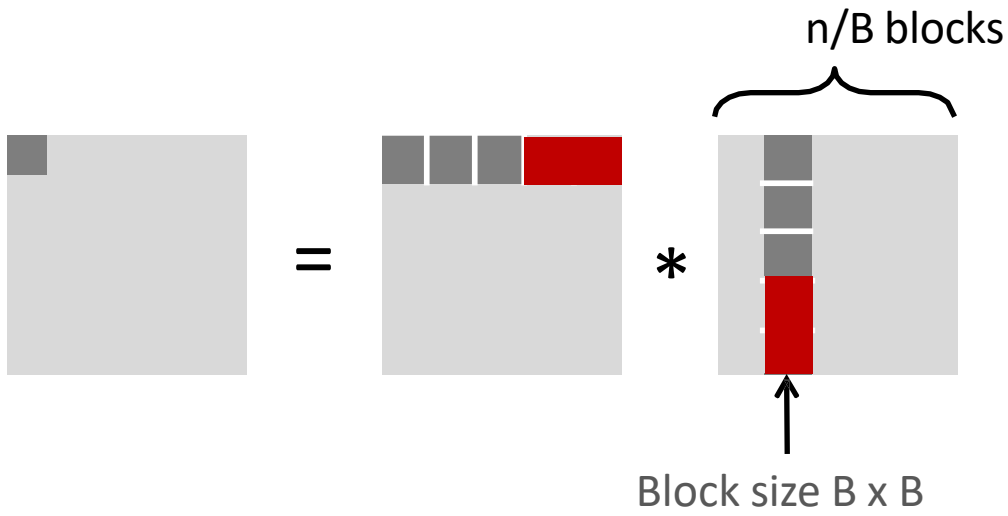
  - Afterwards in cache (schematic)



Cache Miss Analysis

- Assume:
  - Cache block = 8 doubles
  - Cache size << n (much smaller than n)
  - Three blocks ■ fit into cache, i.e., $3B^2 < C$

- Second (block) iteration:
  - Same as first iteration
  - $2 * \frac{n}{B} * \frac{B^2}{8} = \frac{2*nB}{8}$



- Total misses:
  - $\frac{2*nB}{8} * (\frac{n}{B})^2 = \frac{2*n^3}{8*B}$

Summary

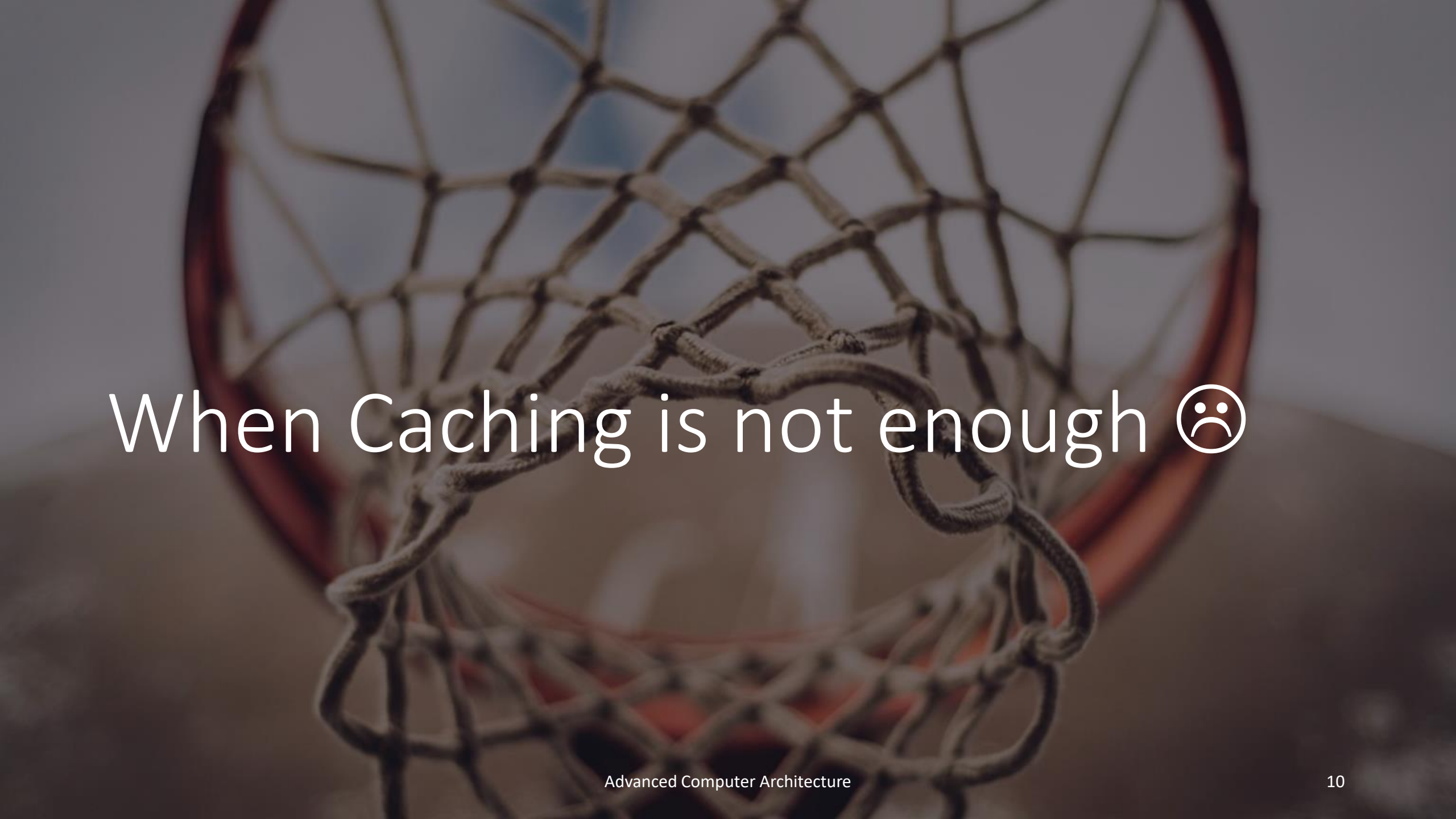
Without blocking	With blocking
$\frac{9}{8} * n^3$	$\frac{2}{8B} * n^3$

- Find largest possible block size BLK, but limit $3BLK^2 < C$!
- Reason for dramatic difference is that matrix multiplication has inherent temporal locality
  - Input data is $3n^2$, computation is $2n^3$, and every array element is used $O(n)$ times!
  - But the program has to be written properly (choose good variant and BLK)

Speedup

If B = 16, 72X speedup ☺ ☺

Why 16?



When Caching is not enough ☹

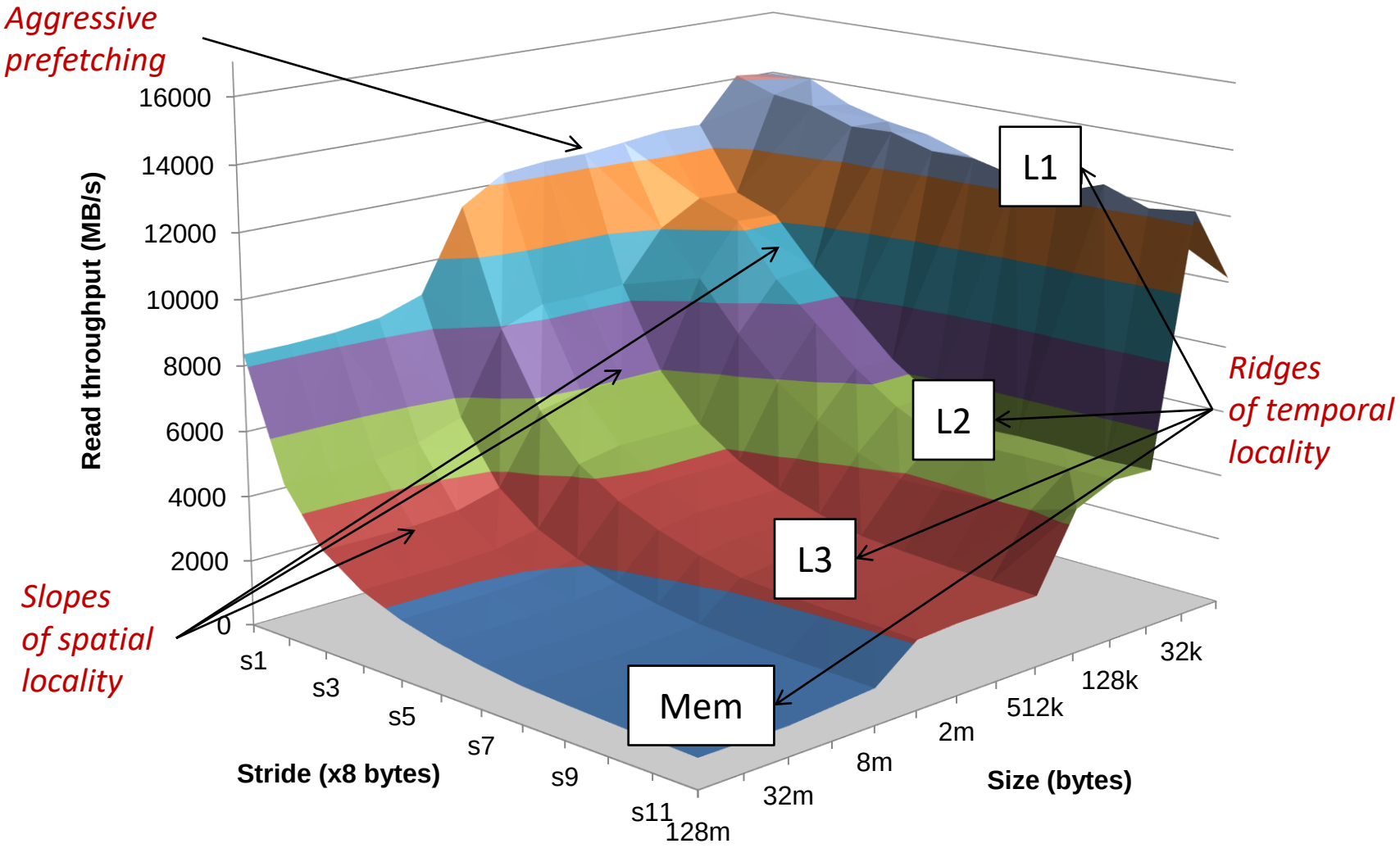
Let’s do software data prefetching (Compiler can do it too)

https://johnnysswlab.com/the-pros-and-cons-of-explicit-software-prefetching/

Try this in gcc and clang

```
for (size_t i = 0; i < n; i++) {
__builtin_prefetch(&a[index[i + 8]]);
  a[index[i]]++;
}
```

Memory Mountain



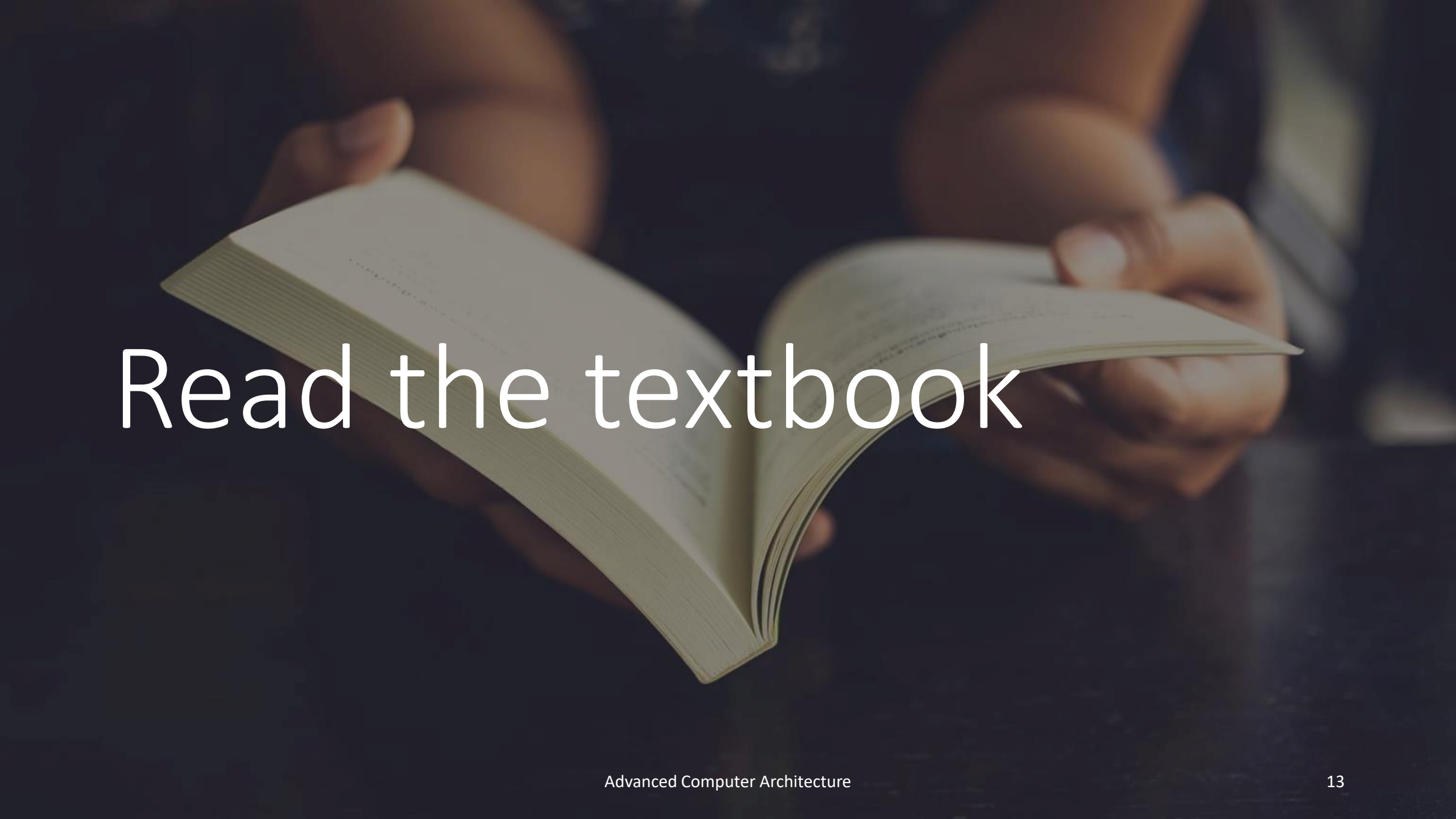
Core i7 Haswell

2.1 GHz
32 KB L1 d-cache

256 KB L2 cache

8 MB L3 cache

64 B block size



Read the textbook

What else ? I still need performance

SIMD magic

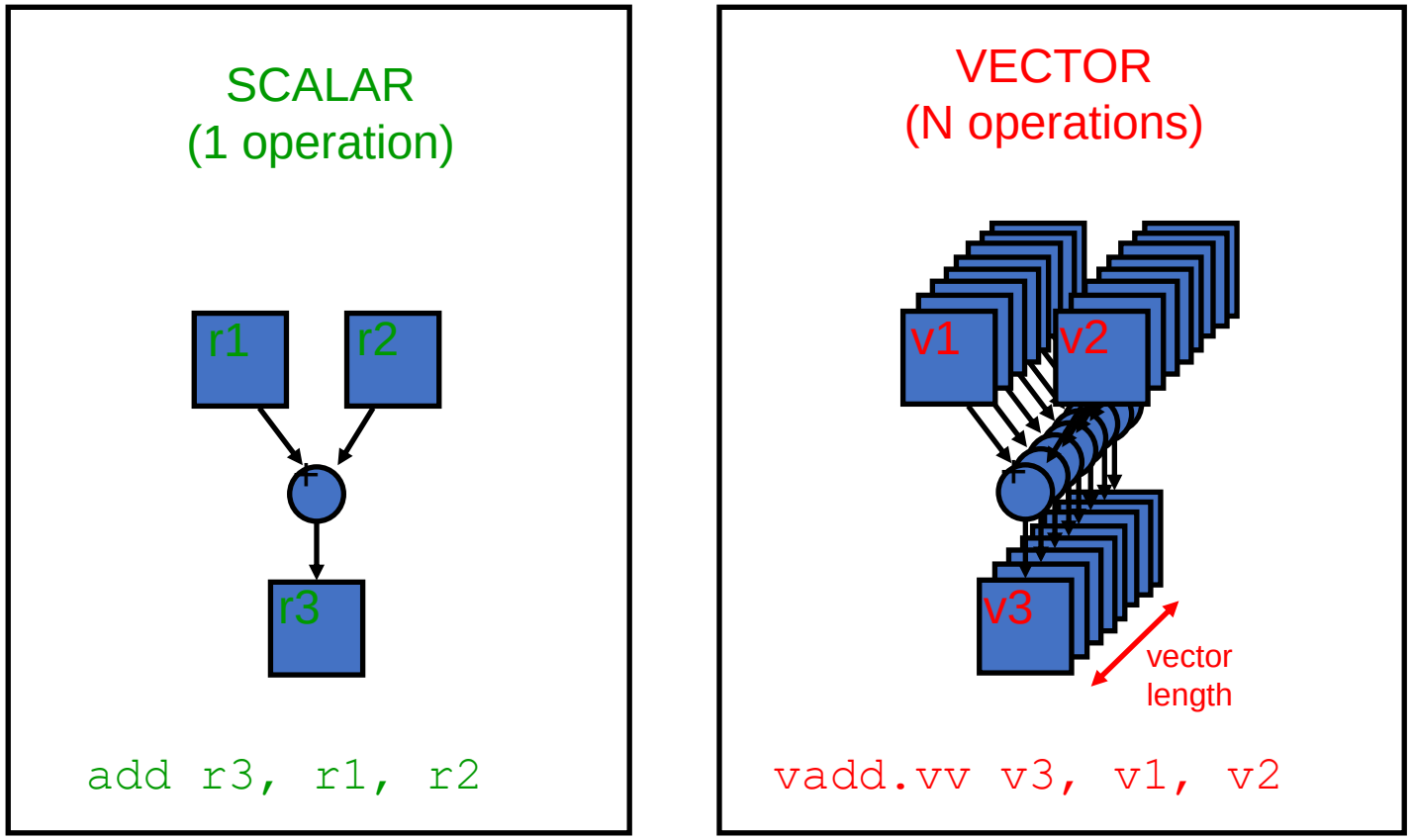
- Single instruction on Multiple Data magic

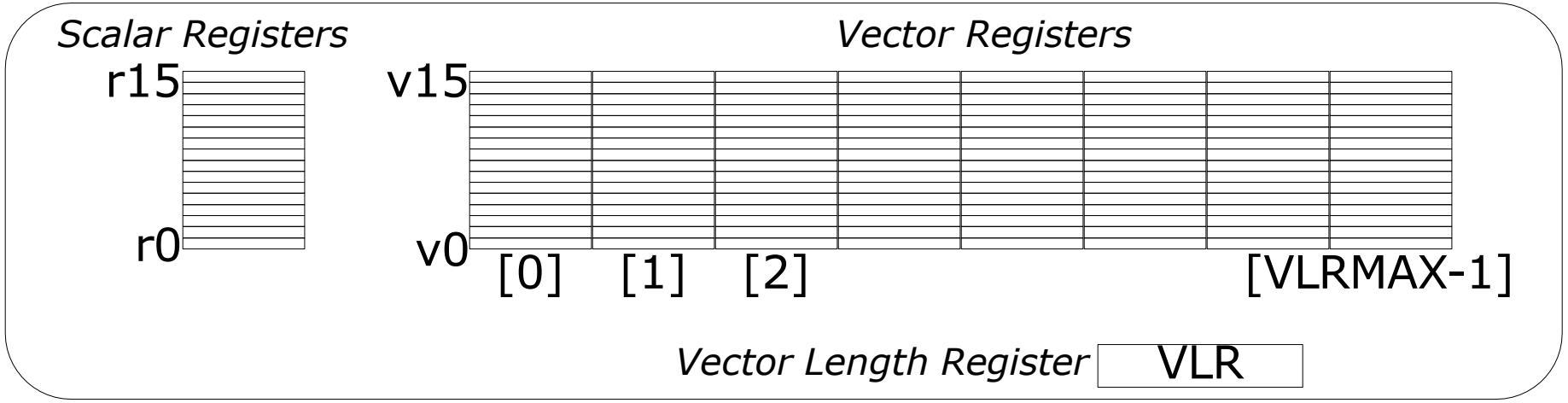
- SIMD is good for applying identical computations across many data elements – E.g.

```
for(i = 0; i < 100; i++) {
A[i] = B[i] + C[i];
}
```

Also “data level parallelism”

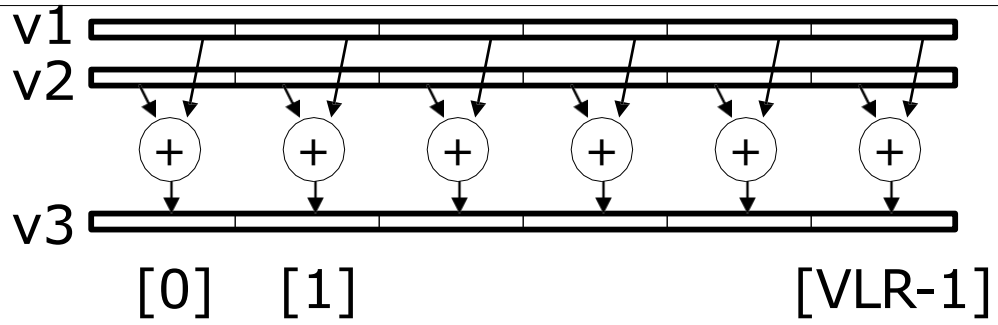
The magic
(Compiler (programmer) ensures no dependencies)





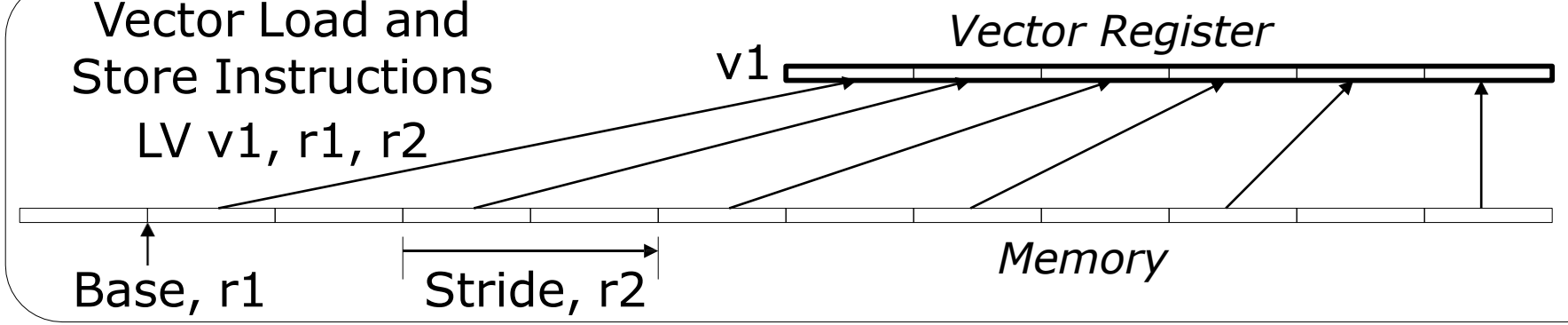
Vector Arithmetic Instructions

ADDV v3, v1, v2

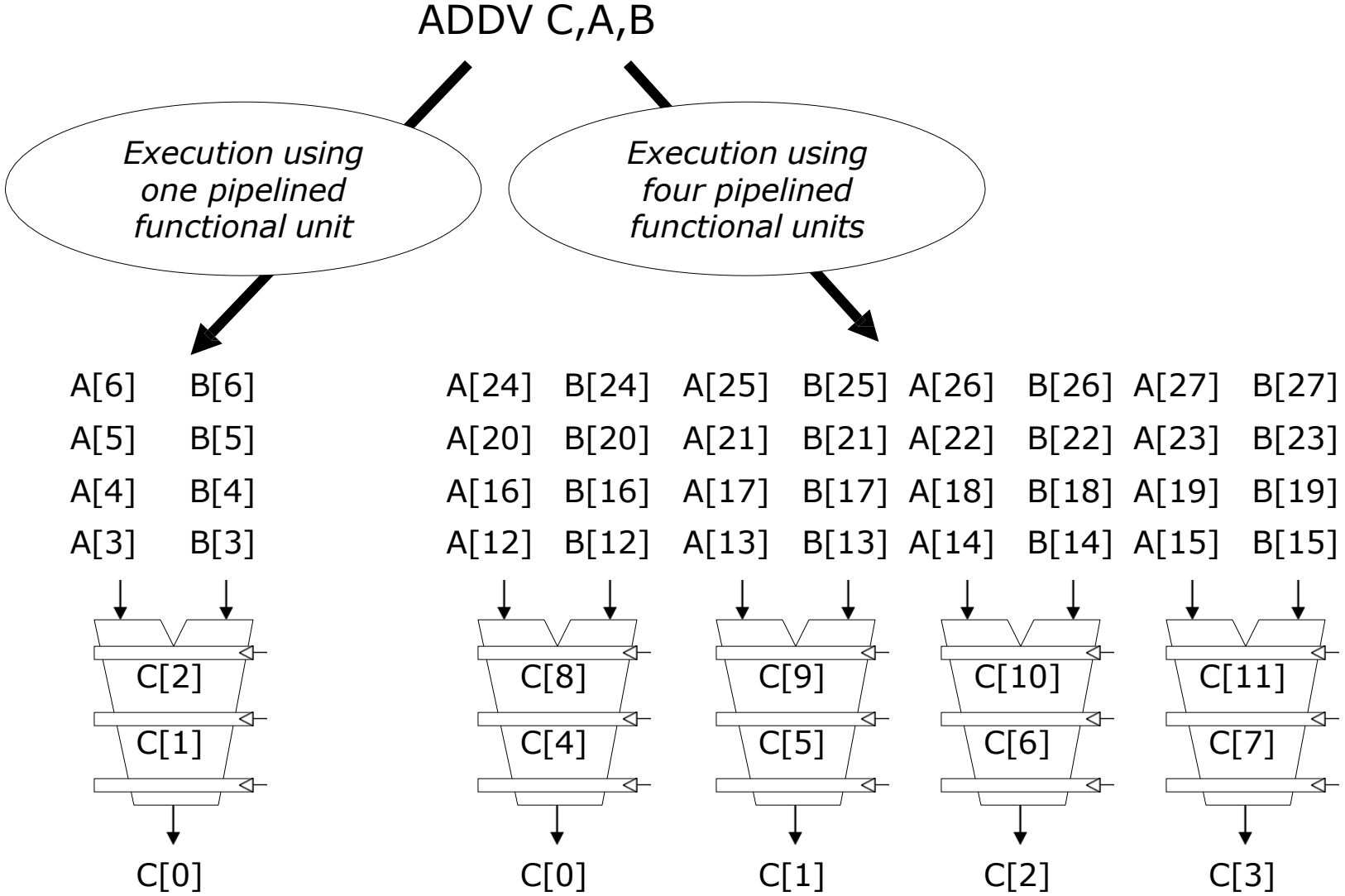


Vector Load and Store Instructions

LV v1, r1, r2



More Execution Units the Better



Look for

- SSE instructions in x86

https://en.wikipedia.org/wiki/Streaming_SIMD_Extensions

Refer Textbook H&P

Programming Assignment-1 Heads-up (coming on 13th August)